

Q1 - 24 January - Shift 1

oscillating at $x = 0$)

The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has:

- (1) exactly two solutions in $(-\infty, \infty)$
- (2) no solution
- (3) a unique solution in $(-\infty, 1)$
- (4) a unique solution in $(-\infty, \infty)$

Space for your notes:

Q2 - 24 January - Shift 2

Let $f(x)$ be a function such that $f(x + y) = f(x) \cdot f(y)$

for all $x, y \in \mathbb{N}$. If $f(1) = 3$ and $\sum_{k=1}^n f(k) = 3279$,

then the value of n is

- (1) 6
- (2) 8
- (3) 7
- (4) 9

Space for your notes:

Q3 - 24 January - Shift 2

If $f(x) = \frac{2^{2x}}{2^{2x} + 2}$, $x \in \mathbb{R}$,

then $f\left(\frac{1}{2023}\right) + f\left(\frac{2}{2023}\right) + \dots + f\left(\frac{2022}{2023}\right)$ is

equal to

- (1) 2011
- (2) 1010
- (3) 2010
- (4) 1011

Space for your notes:

Q4 - 25 January - Shift 1

For some $a, b, c \in \mathbb{N}$, let $f(x) = ax - 3$ and

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$$g(x) = x^b + c, x \in \mathbb{R}. \text{ If } (f \circ g)^{-1}(x) = \left(\frac{x-7}{2} \right)^{1/3}$$

then $(f \circ g)(ac) + (g \circ f)(b)$ is equal to ____.

Q5 - 25 January - Shift 2

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) =$

Space for your notes:

$$\log_{\sqrt{m}} \left\{ \sqrt{2}(\sin x - \cos x) + m - 2 \right\}, \text{ for some } m,$$

such that the range of f is $[0, 2]$. Then the value of

m is

- (1) 5 (2) 3 (3) 2 (4) 4

Q6 - 25 January - Shift 2

The number of functions

Space for your notes:

$$f: \{1, 2, 3, 4\} \rightarrow \{a \in \mathbb{Z} : |a| \leq 8\} \text{ satisfying } f(n) +$$

$$\frac{1}{n} f(n+1) = 1, \forall n \in \{1, 2, 3\} \text{ is}$$

- (1) 3 (2) 4
(3) 1 (4) 2

Q7 - 25 January - Shift 2

Let $f(x) = 2x^n + \lambda, \lambda \in \mathbb{R}, n \in \mathbb{N}$, and $f(4) = 133,$

Space for your notes:

$f(5) = 255$. Then the sum of all the positive integer divisors of $(f(3) - f(2))$ is

- (1) 61 (2) 60
(3) 58 (4) 59

Q8 - 29 January - Shift 1

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

Space for your notes:

$$f(x) = \frac{x^2 + 2x + 1}{x^2 + 1}. \text{ Then}$$

- (1) $f(x)$ is many-one in $(-\infty, -1)$
- (2) $f(x)$ is many-one in $(1, \infty)$
- (3) $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$
- (4) $f(x)$ is one-one in $(-\infty, \infty)$

Q9 - 29 January - Shift 1

The domain of $f(x) = \frac{\log_{(x+1)}(x-2)}{e^{2\log_e x} - (2x+3)}, x \in \mathbb{R}$ is

Space for your notes:

- (1) $\mathbb{R} - \{1-3\}$
- (2) $(2, \infty) - \{3\}$
- (3) $(-1, \infty) - \{3\}$
- (4) $\mathbb{R} - \{3\}$

Q10 - 29 January - Shift 1

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero non-coplanar vectors. Let the position vectors of four points A, B, C and D be $\vec{a} - \vec{b} + \vec{c}$, $\lambda\vec{a} - 3\vec{b} + 4\vec{c}$, $-\vec{a} + 2\vec{b} - 3\vec{c}$ and $2\vec{a} - 4\vec{b} + 6\vec{c}$ respectively. If $\overline{AB}$, $\overline{AC}$ and $\overline{AD}$ are coplanar, then λ is :

Space for your notes:

Q11 - 29 January - Shift 2

Consider a function $f : \mathbb{N} \rightarrow \mathbb{R}$, satisfying

$$f(1) + 2f(2) + 3f(3) + \dots + xf(x) = x(x+1)f(x); x \geq 2$$

with $f(1)=1$. Then $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ is equal to

- (1) 8200
- (2) 8000
- (3) 8400
- (4) 8100

Space for your notes:

Q12 - 30 January - Shift 1

Suppose $f : \mathbb{R} \rightarrow (0, \infty)$ be a differentiable

function such that $5f(x+y) = f(x) \cdot f(y), \forall x, y \in \mathbb{R}$. If

$f(3) = 320$, then $\sum_{n=0}^5 f(n)$ is equal to :

- (1) 6875
- (2) 6575
- (3) 6825
- (4) 6528

Space for your notes:

Q13 - 30 January - Shift 1

Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the number of one-one functions $f : S \rightarrow P(S)$, where $P(S)$ denote the power set of S , such that $f(n) \subset f(m)$ where $n < m$ is _____.

Space for your notes:

Q14 - 30 January - Shift 1

Let $f^1(x) = \frac{3x+2}{2x+3}, x \in \mathbb{R} - \left\{ -\frac{3}{2} \right\}$

For $n \geq 2$, define $f^n(x) = f^1 \circ f^{n-1}(x)$.

If $f^5(x) = \frac{ax+b}{bx+a}, \gcd(a,b)=1$, then $a+b$ is equal to _____.

Space for your notes:

Q15 - 30 January - Shift 2

The range of the function $f(x) = \sqrt{3-x} + \sqrt{2+x}$ is

Space for your notes:

(1) $[\sqrt{5}, \sqrt{10}]$

(2) $[2\sqrt{2}, \sqrt{11}]$

(3) $[\sqrt{5}, \sqrt{13}]$

(4) $[\sqrt{2}, \sqrt{7}]$

Q16 - 30 January - Shift 2

Let $A = \{1, 2, 3, 5, 8, 9\}$. Then the number of possible functions $f: A \rightarrow A$ such that $f(m \cdot n) = f(m) \cdot f(n)$ for every $m, n \in A$ with $m \cdot n \in A$ is equal to _____.

Space for your notes:

Q17 - 31 January - Shift 1

If the domain of the function $f(x) = \frac{[x]}{1+x^2}$, where

$[x]$ is greatest integer $\leq x$, is $(2, 6)$, then its range is

(1) $\left(\frac{5}{26}, \frac{2}{5}\right] - \left\{\frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53}\right\}$

(2) $\left(\frac{5}{26}, \frac{2}{5}\right]$

(3) $\left(\frac{5}{37}, \frac{2}{5}\right] - \left\{\frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53}\right\}$

(4) $\left(\frac{5}{37}, \frac{2}{5}\right]$

Space for your notes:

Q18 - 31 January - Shift 2

The absolute minimum value, of the function

$f(x) = |x^2 - x + 1| + [x^2 - x + 1]$, where $[t]$

denotes the greatest integer function, in the interval

$[-1, 2]$, is :

(1) $\frac{3}{4}$

(2) $\frac{3}{2}$

(3) $\frac{1}{4}$

(4) $\frac{5}{4}$

Space for your notes:

Q19 - 31 January - Shift 2

Let $f: \mathbb{R} - \{2, 6\} \rightarrow \mathbb{R}$ be real valued function

Space for your notes:

defined as $f(x) = \frac{x^2 + 2x + 1}{x^2 - 8x + 12}$. Then range of f is

(1) $\left(-\infty, -\frac{21}{4}\right] \cup [0, \infty)$

(2) $\left(-\infty, -\frac{21}{4}\right) \cup (0, \infty)$

(3) $\left(-\infty, -\frac{21}{4}\right] \cup \left[\frac{21}{4}, \infty\right)$

(4) $\left(-\infty, -\frac{21}{4}\right] \cup [1, \infty)$

Q20 - 01 February - Shift 2

Let $f: \mathbb{R} - \{0, 1\} \rightarrow \mathbb{R}$ be a function such that

Space for your notes:

$f(x) + f\left(\frac{1}{1-x}\right) = 1 + x$. Then $f(2)$ is equal to :

(1) $\frac{9}{2}$

(2) $\frac{9}{4}$

(3) $\frac{7}{4}$

(4) $\frac{7}{3}$

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (4)	Q2 (3)	Q3 (4)	Q4 (2039)
Q5 (1)	Q6 (4)	Q7 (2)	Q8 (3)
Q9 (2)	Q10 (2)	Q11 (4)	Q12 (3)
Q13 (3240)	Q14 (3125)	Q15 (1)	Q16 (432)
Q17 (4)	Q18 (1)	Q19 (1)	Q20 (2)

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Q1 (4)

$$x^2 - 4x + [x] + 3 = x[x]$$

$$\Rightarrow x^2 - 4x + 3 = x[x] - [x]$$

$$\Rightarrow (x-1)(x-3) = [x] \cdot (x-1)$$

$$\Rightarrow x = 1 \text{ or } x - 3 = [x]$$

$$\Rightarrow x - [x] = 3$$

$$\Rightarrow \{x\} = 3 \text{ (Not Possible)}$$

Only one solution $x = 1$ in $(-\infty, \infty)$

Q2 (3)

$$f(x + y) = f(x) \cdot f(y) \quad \forall x, y \in \mathbb{N}, f(1) = 3$$

$$f(2) = f^2(1) = 3^2$$

$$f(3) = f(1) f(2) = 3^3$$

$$f(4) = 3^4$$

$$f(k) = 3^k$$

$$\sum_{k=1}^n f(k) = 3279$$

$$f(1) + f(2) + f(3) + \dots + f(k) = 3279$$

$$3 + 3^2 + 3^3 + \dots + 3^k = 3279$$

$$\frac{3(3^k - 1)}{3 - 1} = 3279$$

$$\frac{3^k - 1}{2} = 1093$$

$$3^k - 1 = 2186$$

$$3^k = 2187$$

$$\boxed{k = 7}$$

Q3 (4)

$$f(x) = \frac{4^x}{4^x + 2}$$

$$f(x) + f(1-x) = \frac{4^x}{4^x + 2} + \frac{4^{1-x}}{4^{1-x} + 2}$$

$$= \frac{4^x}{4^x + 2} + \frac{4}{4 + 2(4^x)}$$

$$= \frac{4^x}{4^x + 2} + \frac{2}{2 + 4^x}$$

$$= 1$$

$$\Rightarrow f(x) + f(1-x) = 1$$

$$\text{Now } f\left(\frac{1}{2023}\right) + f\left(\frac{2}{2023}\right) + f\left(\frac{3}{2023}\right) + \dots +$$

$$\dots + f\left(1 - \frac{3}{2023}\right) + f\left(1 - \frac{2}{2023}\right) + f\left(1 - \frac{1}{2023}\right)$$

Now sum of terms equidistant from beginning and

end is 1

$$\text{Sum} = 1 + 1 + 1 + \dots + 1 \text{ (1011 times)}$$

$$= 1011$$

Q4 (2039)

$$\text{Let } fog(x) = h(x)$$

$$\Rightarrow h^{-1}(x) = \left(\frac{x-7}{2} \right)^{\frac{1}{3}}$$

$$\Rightarrow h(x) = fog(x) = 2x^3 + 7$$

$$fog(x) = a(x^b + c) - 3$$

$$\Rightarrow a = 2, b = 3, c = 5$$

$$\Rightarrow fog(ac) = fog(10) = 2007$$

$$g(f(x)) = (2x - 3)^3 + 5$$

$$\Rightarrow gof(b) = gof(3) = 32$$

$$\Rightarrow \text{sum} = 2039$$

Q5 (1)

Since,

$$-\sqrt{2} \leq \sin x - \cos x \leq \sqrt{2}$$

$$\therefore -2 \leq \sqrt{2} (\sin x - \cos x) \leq 2$$

$$\text{(Assume } \sqrt{2} (\sin x - \cos x) = k \text{)}$$

$$-2 \leq k \leq 2 \quad \dots \text{(i)}$$

$$f(x) = \log_{\sqrt{m}} (k + m - 2)$$

Given,

$$0 \leq f(x) \leq 2$$

$$0 \leq \log_{\sqrt{m}} (k + m - 2) \leq 2$$

$$1 \leq k + m - 2 \leq m$$

$$-m + 3 \leq k \leq 2 \quad \dots \text{(ii)}$$

From eq. (i) & (ii), we get $-m + 3 = -2$

$$\Rightarrow m = 5$$

Q6 (4)

$$f: \{1, 2, 3, 4\} \rightarrow \{a \in \mathbb{Z} : |a| \leq 8\}$$

$$f(n) + \frac{1}{n}f(n+1) = 1, \forall n \in \{1, 2, 3\}$$

$f(n+1)$ must be divisible by n

$$f(4) \Rightarrow -6, -3, 0, 3, 6$$

$$f(3) \Rightarrow -8, -6, -4, -2, 0, 2, 4, 6, 8$$

$$f(2) \Rightarrow -8, \dots, 8$$

$$f(1) \Rightarrow -8, \dots, 8$$

$\frac{f(4)}{3}$ must be odd since $f(3)$ should be even

therefore 2 solution possible.

$$\begin{matrix} f(4) & f(3) & f(2) & f(1) \end{matrix}$$

$$\begin{matrix} -3 & 2 & 0 & 1 \end{matrix}$$

$$\begin{matrix} 3 & 0 & 1 & 0 \end{matrix}$$

Q7 (2)

$$f(x) = 2x^n + \lambda$$

$$f(4) = 133$$

$$f(5) = 255$$

$$133 = 2 \times 4^n + \lambda \quad (1)$$

$$255 = 2 \times 5^n + \lambda \quad (2)$$

$$(2) - (1)$$

$$122 = 2(5^n - 4^n)$$

$$\Rightarrow 5^n - 4^n = 61$$

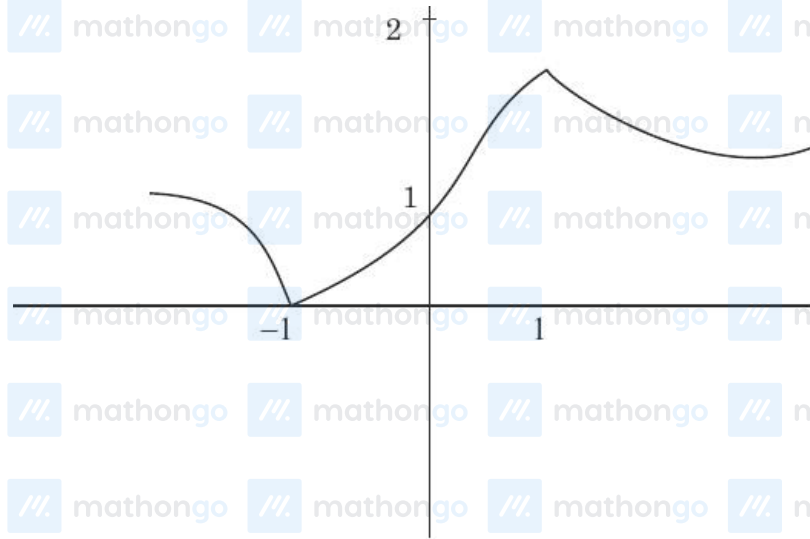
$$\therefore n = 3 \text{ \& } \lambda = 5$$

$$\text{Now, } f(3) - f(2) = 2(3^3 - 2^3) = 38$$

Number of Divisors is 1, 2, 19, 38 ;

their sum is 60

Q8 (3)



$$f(x) = \frac{(x+1)^2}{x^2+1} = 1 + \frac{2x}{x^2+1}$$

$$f(x) = 1 + \frac{2}{x + \frac{1}{x}}$$

Q9 (2)

$$x - 2 > 0 \Rightarrow x > 2$$

$$x + 1 > 0 \Rightarrow x > -1$$

$$x + 1 \neq 1 \Rightarrow x \neq 0 \text{ and } x > 0$$

Denominator

$$x^2 - 2x - 3 \neq 0$$

$$(x - 3)(x + 1) \neq 0$$

$$x \neq -1, 3$$

So Ans $(2, \infty) - \{3\}$

Q10 (2)

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$$\overline{AB} = (\lambda - 1)\overline{a} - 2\overline{b} + 3\overline{c}$$

$$\overline{AC} = 2\overline{a} + 3\overline{b} - 4\overline{c}$$

$$\overline{AD} = \overline{a} - 3\overline{b} + 5\overline{c}$$

$$\begin{vmatrix} \lambda - 1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (\lambda - 1)(15 - 12) + 2(-10 + 4) + 3(6 - 3) = 0$$

Q11 (4)

Given for $x \geq 2$

$$f(1) + 2f(2) + \dots + xf(x) = x(x + 1) f(x)$$

replace x by $x + 1$

$$\begin{aligned} \Rightarrow x(x + 1) f(x) + (x + 1) f(x + 1) \\ = (x + 1)(x + 2) f(x + 1) \end{aligned}$$

$$\Rightarrow \frac{x}{f(x + 1)} + \frac{1}{f(x)} = \frac{(x + 2)}{f(x)}$$

$$\Rightarrow x f(x) = (x + 1) f(x + 1) = \frac{1}{2}, x \geq 2$$

$$f(2) = \frac{1}{4}, f(3) = \frac{1}{6}$$

$$\text{Now } f(2022) = \frac{1}{4044}$$

$$f(2028) = \frac{1}{4056}$$

$$\text{So, } \frac{1}{f(2022)} + \frac{1}{f(2028)} = 4044 + 4056 = 8100$$

Q12 (3)

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Option (3)

$$5f(x+y) = f(x) \cdot f(y)$$

$$5f(0) = f(0)^2 \Rightarrow f(0) = 5$$

$$5f(x+1) = f(x) \cdot f(1)$$

$$\Rightarrow \frac{f(x+1)}{f(x)} = \frac{f(1)}{5}$$

$$\Rightarrow \frac{f(1)}{f(0)} \cdot \frac{f(2)}{f(1)} \cdot \frac{f(3)}{f(2)} = \left(\frac{f(1)}{5} \right)^3$$

$$\Rightarrow \frac{320}{5} = \frac{(f(1))^3}{5^3} \Rightarrow f(1) = 20$$

$$\therefore 5f(x+1) = 20 \cdot f(x) \Rightarrow f(x+1) = 4f(x)$$

$$\sum_{n=0}^5 f(n) = 5 + 5.4 + 5.4^2 + 5.4^3 + 5.4^4 + 5.4^5$$

$$= \frac{5[4^6 - 1]}{3} = 6825$$

Q13 (3240)

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Let $S = \{1, 2, 3, 4, 5, 6\}$, then the number of one-one functions, $f: S \rightarrow P(S)$, where $P(S)$ denotes the power set of S , such that $f(n) \subset f(m)$ where $n < m$ is

$$n(S) = 6$$

$$P(S) = \left\{ \phi, \{1\}, \dots, \{6\}, \{1, 2\}, \dots, \{5, 6\}, \dots, \{1, 2, 3, 4, 5, 6\} \right\}$$

– 64 elements

case – 1

$f(6) = S$ i.e. 1 option,

$f(5) =$ any 5 element subset A of S i.e. 6 options,

$f(4) =$ any 4 element subset B of A i.e. 5 options,

$f(3) =$ any 3 element subset C of B i.e. 4 options,

$f(2) =$ any 2 element subset D of C i.e. 3 options,

$f(1) =$ any 1 element subset E of D or empty subset i.e. 3 options,

Total functions = 1080

Case – 2

$f(6) =$ any 5 element subset A of S i.e. 6 options,

$f(5) =$ any 4 element subset B of A i.e. 5 options,

$f(4) =$ any 3 element subset C of B i.e. 4 options,

$f(3) =$ any 2 element subset D of C i.e. 3 options,

$f(2) =$ any 1 element subset E of D i.e. 2 options,

$f(1) =$ empty subset i.e. 1 option

Total functions = 720

Case – 3

$f(6) = S$

$f(5) =$ any 4 element subset A of S i.e. 15 options,

$f(4) =$ any 3 element subset B of A i.e. 4 options,

$f(3) =$ any 2 element subset C of B i.e. 3 options,

$f(2) =$ any 1 element subset D of C i.e. 2 options,

$f(1) =$ empty subset i.e. 1 option

Total functions = 360

Case – 4

$f(6) = S$

$f(5) =$ any 5 element subset A of S i.e. 6 options,

$f(4) =$ any 3 element subset B of A i.e. 10 options,

$f(3) =$ any 2 element subset C of B i.e. 3 options,

$f(2) =$ any 1 element subset D of C i.e. 2 options,

$f(1) =$ empty subset i.e. 1 option

Total functions = 360

Case – 5

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Questions with Solutions

MathonGo

$f(5)$ = any 5 element subset A of S i.e. 6 options,
 $f(4)$ = any 4 element subset B of A i.e. 5 options,
 $f(3)$ = any 2 element subset C of B i.e. 6 options,
 $f(2)$ = any 1 element subset D of C i.e. 2 options

$$f^1(x) = \frac{3x+2}{2x+3}$$

$$\Rightarrow f^2(x) = \frac{13x+12}{12x+13}$$

$$\Rightarrow f^3(x) = \frac{63x+62}{62x+63}$$

$$\therefore f^5(x) = \frac{1563x+1562}{1562x+1563}$$

$$a+b=3125$$

Q15 (1)

$$y^2 = 3-x+2+x+2\sqrt{(3-x)(2+x)}$$

$$= 5+2\sqrt{6+x-x^2}$$

$$y^2 = 5+2\sqrt{\frac{25}{4}-\left(x-\frac{1}{2}\right)^2}$$

$$y_{\max} = \sqrt{5+5} = \sqrt{10}$$

$$y_{\min} = \sqrt{5}$$

Q16 (432)

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$$f(1) = 1; f(9) = f(3) \times f(3)$$

$$\text{i.e., } f(3) = 1 \text{ or } 3$$

$$\text{Total function} = 1 \times 6 \times 2 \times 6 \times 6 \times 1 = 432$$

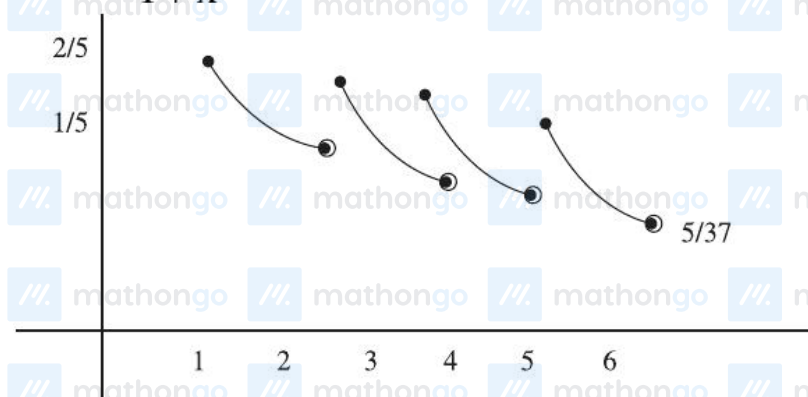
Q17 (4)

$$f(x) = \frac{2}{1+x^2} \quad x \in [2, 3)$$

$$f(x) = \frac{3}{1+x^2} \quad x \in [3, 4)$$

$$f(x) = \frac{4}{1+x^2} \quad x \in [4, 5)$$

$$f(x) = \frac{5}{1+x^2} \quad x \in [5, 6)$$



$$\left(\frac{5}{37}, \frac{2}{5} \right]$$

Q18 (1)

$$f(x) = |x^2 - x + 1| + [x^2 - x + 1]; x \in [-1, 2]$$

$$\text{Let } g(x) = x^2 - x + 1$$

$$= \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\therefore |x^2 - x + 1| \text{ and } [x^2 - x + 1]$$

Both have minimum value at $x = 1/2$

$$\Rightarrow \text{Minimum } f(x) = \frac{3}{4} + 0$$

$$= \frac{3}{4}$$

Q19 (1)

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$$\text{Let } y = \frac{x^2 + 2x + 1}{x^2 - 8x + 12}$$

By cross multiplying

$$yx^2 - 8xy + 12y - x^2 - 2x - 1 = 0$$

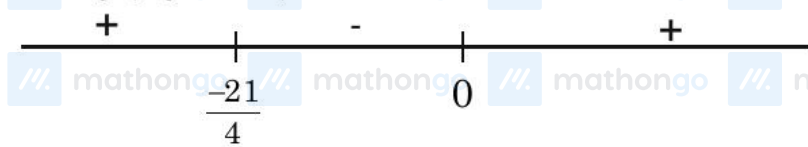
$$x^2(y - 1) - x(8y + 2) + (12y - 1) = 0$$

Case 1, $y \neq 1$

$$D \geq 0$$

$$\Rightarrow (8y + 2)^2 - 4(y - 1)(12y - 1) \geq 0$$

$$\Rightarrow y(4y + 21) \geq 0$$



$$y \in \left(-\infty, -\frac{21}{4}\right] \cup [0, \infty) - \{1\}$$

Case 2, $y = 1$

$$x^2 + 2x + 1 = x^2 - 8x + 12$$

$$10x = 11$$

$$x = \frac{11}{10}$$

So, y can be 1

Q20 (2)

$$f(x) + f\left(\frac{1}{1-x}\right) = 1 + x$$

$$x = 2 \Rightarrow f(2) + f(-1) = 3 \quad (1)$$

$$x = -1 \Rightarrow f(-1) + f\left(\frac{1}{2}\right) = 0 \quad (2)$$

$$x = \frac{1}{2} \Rightarrow f\left(\frac{1}{2}\right) + f(2) = \frac{3}{2} \quad (3)$$

$$(1) + (3) - (2) \Rightarrow 2f(2) = \frac{9}{2}$$

$$\therefore f(2) = \frac{9}{4}$$